



Over the duration of the last one year, NVMe SSDs haven't managed to become as economical as expected.

In fact, the prices of the SSDs have more or less remained stagnant on a cost/GB basis. What has changed, however, is that more and more SSD manufacturers have started improving upon the controllers. The average NVMe speed has gone up on the consumer SSD front as well as on the OEM front, so most of the SSDs that we see bundled in laptops are also consistently improving. As for the NAND technology, it definitely seems to be on the upswing as well. Chinese silicon manufacturer, YMTC has promised to ship 64-layer 3D NAND in 2019 with support for Xtacing technology. This might give us NAND flash with I/O speeds that would rival that of flash memory. So we could see ultra-fast NAND tech. However, it would seem that the only bottleneck from that point onwards will be the interface bandwidth. So 2018 has been not so memorable but 2019 might be an entirely exciting year. Let's take a look at our winner of the Zero1 award in this category of the Best NVMe SSD.

NVMe SSD

HITTING THE LIMIT



WD Black 1TB 2018

Price: ₹32,999



Corsair Force MP510

Price: ₹28,500



Kingston A1000

Price: ₹4,950

2018 ZERO1 AWARD WINNER WD Black 1TB 2018

The new WD Black NVMe SSD is definitely a really good SSD. It has successfully managed to best some of the enthusiast favourites in quite a lot of the benchmarks. The Intel 760p and the Samsung 960 Pro did end up being a bit competitive, however, at the end, the WD Black NVMe turned out to be the better drive. The controller on the new WD Black NVMe SSD was designed internally and most of the specs are still held in complete mystery. Considering that M.2 PCIe x4 slots have a theoretical speed limit of about 3940 MB/s, the transfer speeds which averaged at about 3400 MB/s appear to be quite a massive improvement over the older Black SSD. For the price of 32,999, the cost per GB comes down to ₹33 which is quite economical as well. Let's just say that WD's efforts with the new controller and NAND have finally worked their magic which has granted them the winner's spot in the category of the Best NVMe SSDs.

RUNNER-UP Corsair Force MP510

An upgrade to last year's MP500, is the new Corsair Force MP510, comes with the Toshiba BiCS3 64-layer 3D TLC NAND with a Phison PS5012-E12 for the controller. This gives the MP510 consistent and high read speeds but not so great write speeds, unfortunately. Moreover, if we take a look at the IOPS figures for the Corsair MP510 then we notice that there's a significant improvement across the board. The MP510 loses out to the WD Black primarily because of the write speeds but it's a great drive nevertheless, therefore placing it at the runner-up spot in this category.

BEST BUY Kingston A1000

The Kingston A1000 came out early this year and has had a significant price reduction over the year. It offers both, a mix of good performance as well as low cost/GB. Like the MP510, the A1000 has the same 64-layer Toshiba BiCS3 NAND but the controller is the Phison PS5008-E8 which is a slightly scaled down version of the one in the MP510. As a result, we see read speeds of about 1,500 MB/s but write speeds hover in the 800 MB/s range. Compared to the other drives we saw this year, the A1000 has a cost/GB of ₹20.6 which makes it an ideal buy for folks wanting to jump onto the NVMe bandwagon. This is why, the Kingston A1000 is our Best Buy for 2018.